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1 Abstract

This study aims to determine the evolution of key quantities such as the mixed-layer depth or the salinity in a saline convection tank experiment of entrainment fluxes to better understand mixed-layer dynamics. Results indicate that the mixed-layer salinity decreases over time, leading to a reduction in surface buoyancy flux. However, the convective velocity scale remains constant due to the product of mixed-layer depth and buoyancy flux being constant. The entrainment fluxes were measured using dye concentration techniques, revealing that the actual entrainment ratios (A) were significantly lower than anticipated. First approximation using an entrainment ratio of 0.2 showed decreasing fluorescein concentrations, prompting adjustments to lower presumed ratios. More experiments with $A = 0.1$ and $A = 0.05$ were carried out and also demonstrated a lower entrainment than expected. The measured fluorescein concentrations suggest an entrainment ratio close to 0.02, way smaller than reported values for similar setups. Additional validation tests such as the study of the stability of the fluorescence dye or the direct determination of the entrainment velocity confirmed the accuracy of the measurement methods. Furthermore, the conclusion that the entrainment flux in this saline convection setup is an order of magnitude lower than expected for penetrative convection is supported.